A primer on structured decision making for invasive species (using dynamic programming)

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Invasive species (IAS) pose significant ecological, economic, and public health challenges. Robust methods are needed for guiding effective management. Overall, we hope to demonstrate the flexibility and utility of structured decision making in invasive species management.

Target journal: Biological invasions? NeoBiota?.

Keywords: Adaptive management, Invasive species, Stochastic dynamic programming, Structured decision making

Introduction

Invasive species are a significant global issue, with wide-ranging impacts on ecosystems, economies, and public health (Pyšek et al. 2020, Roy et al. 2024).

Effective management of invasive species requires blabla for guiding control efforts and evaluating the success of eradication or regulation programs (Williams et al. 2002, Thompson et al. 2021).

Pitch: Structured decision-making (SDM) and adaptive management (AM) have been repeatedly recommended for the management of invasive species, yet they remain underused in practice. There are many reasons for this gap, one of which is the technical challenge of formally identifying and quantifying trade-offs among management objectives. Here, we contribute to ongoing efforts by illustrating the use of stochastic dynamic programming (SDP) and some of its variants — including active and passive adaptive management and partially observable Markov decision processes (POMDPs) — across different ecological contexts (a single population, a metapopulation, and two interacting populations).

Several papers have recommended or implemented structured decision-making (SDM) for managing invasive species. It's worth providing a short review of this literature.

Coypu (Myocastor coypu) as a case study throughout the paper. Code and tutorials available as appendices.

Add a shiny for each case study so that folks can play around?

A primer on structured decision making

Why and what (all)

Stochastic dynamic programming (Olivier)

Example on computer maintenance (coverted into coypu example) to explain SDP on a simple example ?

Case studies with coypus

Single population (Olivier)

Take wolf example and turn it into a coypu example ?

Two interacting species (Lucile)

Competition or predator-prey, or host-pathogen / disease thing. See SDP-species-interactions/ in Olivier biblio repo.

Metapopulation (Olivier)

Acounting for uncertainty

POMDP (Cassie)

Illustrate POMDP, Olivier has some code extending the computer maintenance example from above.

https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14374

https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13692

Adaptive management (Abby)

When is it useful. Start with value of information?

On a single species ; check out Conroy's book, there is a nice figure illustrating AM.

https://onlinelibrary.wiley.com/doi/10.1002/ece3.71176

Discussion (all)

Main messages

How is SDM relevant.

Limitations

Curse of dimensionality.

Perspectives

Acknowledgments

Data availability statement

Data and code are available at https://github.com/oliviergimenez/SDPaper.

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